

Branch Specialization Checkpoint

Two-Attractor Layout Test

Branch Specialization Project

2026-05-22

Checkpoint reason: negative result separates specialization from modularity.

Current mechanism: heterogeneous head dimensions create capacity-attractor slots.

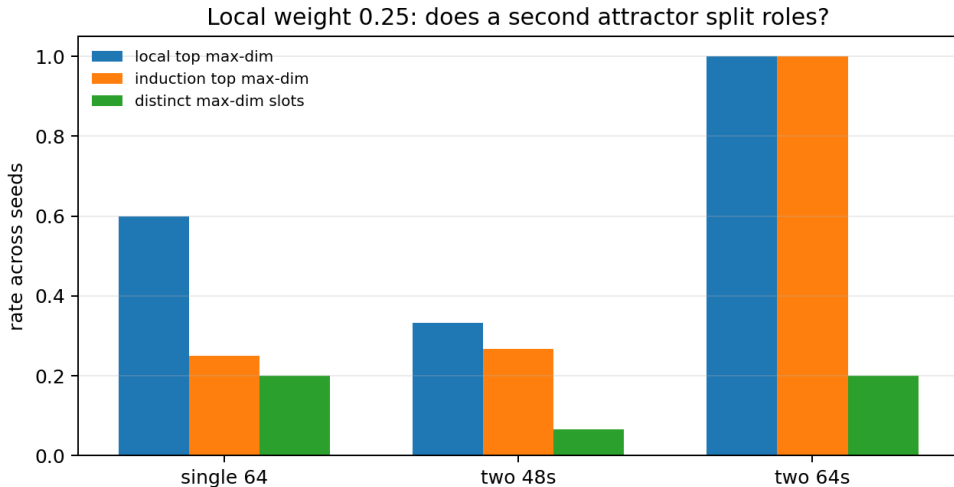
The next implication was:

If one high-capacity slot causes role competition, then two high-capacity slots should let local and induction split more cleanly.

This checkpoint tests that implication under the strongest competition setting: local weight 0.25, induction weight 1.0.

- Same local-vs-induction toy task.
- Role score: single-head causal ablation loss delta.
- Two-48 layouts: [16,48,48,16], [48,16,48,16], [48,48,16,16].
- Control: `uniform2 = [64,64]`.
- Comparison baseline: prior single-64 heterogeneous layouts at local weight 0.25.
- Five seeds per new config.

Main Evidence



The two-48 layouts did not make local and induction split between max-dim heads.

Summary Table

Family	Models	Local top max	Induction top max	Same slot	Distinct max slots
Single 64 hetero	20	0.60	0.25	0.45	0.20
Two 48 hetero	15	0.33	0.27	0.80	0.07
Two 64 uniform	5	1.00	1.00	0.80	0.20

For two-48 layouts, local top heads were 48-dim in only 5/15 models and induction top heads were 48-dim in only 4/15 models.

Supported:

- Adding more capacity does not automatically produce modular role separation.
- Local and induction can cohabit the same top causal slot.
- Differentiated capacity is insufficient evidence for functional modularity.

Not supported:

- Two medium-large heads are enough to make local and induction split roles.
- Two 64-dim heads necessarily split roles; `uniform2` shared the same top slot in 4/5 seeds.

The Stage 3 claim should now be:

Structural heterogeneity can stabilize functional specialization slots, but functional modularity requires more than differentiated head capacity.

This sharpens the central distinction:

specialization does not imply modularity

Continue, but shift the next test from capacity heterogeneity to structural separation.

Recommended next tests:

- Add explicit branch isolation or routing and compare against head-dimension heterogeneity.
- Use a more conflict-heavy two-role task where one shared head is less likely to solve both roles.
- Add multi-head ablation or path patching to distinguish role cohabitation from redundant separate circuits.

Sources and Provenance

- Report: `doc/phase3_toy_competition_two_attractor.md`
- Training script: `scripts/toy_competition_head_dim_intervention.py`
- Analysis script: `scripts/analyze_competition_two_attractor.py`
- New outputs: `results/phase3_toy_competition_two48_lw025/`
- Control output: `results/phase3_toy_competition_uniform2_lw025/`
- Combined analysis: `results/phase3_toy_competition_two_attractor/`